

PROGRAM-4

Integrate Git with Docker for source controlled Application builds

Project Structure:

Dockerfile

[index.js](#)

package.json

package-lock.json

Steps 1: Create a folder prog-4 and navigate to it

Steps 2: Create a package. json file using command npm init -y

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ mkdir p-4
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ cd p-4
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-4$ npm init -y
Wrote to /home/_1RV24MC019_anusha_s/p-4/package.json:

{
  "name": "p-4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

Step 3: Create a [index.js](#) file

```
const express=require("express");
const app=express();
const PORT=3000;

app.get("/",(req,res)=>{
res.send("Hello from Git + Docker integration");
});

app.listen(PORT,()=>{
console.log(`Server is running on http://localhost:${PORT}`);
});
```

Step 4:Add the dependencies

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-4$ npm install express

added 68 packages, and audited 69 packages in 3s

18 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Step 5:Create a Dockerfile and write

```
#Stage-1 Build Stage
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

#Stage-2 Runtime Stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 5000
CMD ["node","index.js"]
```

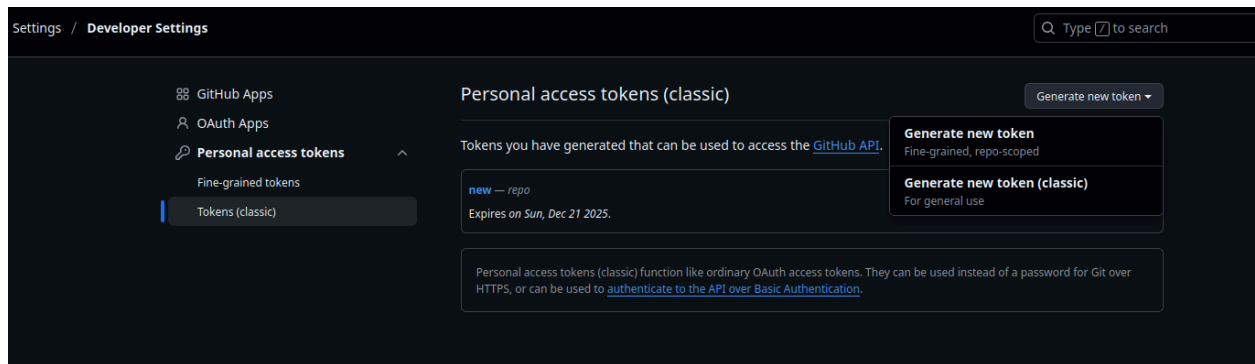
Step 6: Login to your github account and Create a new Repository

Login->New->New Repository->Provide the repository name

Step 7: Generate the personal access token

Go to profile->Settings->Developer Settings->Personal Access Token->Token(classic)->Generate Token

Save it somewhere since it is visible for some time only



Step 8: Initialize the git repository locally

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/_1RV24MC019_anusha_s/p-4/.git/
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-4$ git add .
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-4$ git commit -m "Initial commit"
[master (root-commit) af3b8e5] Initial commit
636 files changed, 72753 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/HISTORY.md
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
create mode 100644 node_modules/body-parser/lib/utils.js
create mode 100644 node_modules/body-parser/package.json
create mode 100644 node_modules/bytes/History.md
create mode 100644 node_modules/bytes/LICENSE
create mode 100644 node_modules/bytes/Readme.md
create mode 100644 node_modules/bytes/index.js
create mode 100644 node_modules/bytes/package.json
create mode 100644 node_modules/call-bind-apply-helpers/.eslintrc
create mode 100644 node_modules/call-bind-apply-helpers/.github/FUNDING.yml
create mode 100644 node_modules/call-bind-apply-helpers/.nycrc
create mode 100644 node_modules/call-bind-apply-helpers/CHANGELOG.md

```

Step 9: Connect to the remote repository

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-4$ git branch -M main
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-4$ git remote add origin https://github.com/A2003nu/prog-4.git

```

Step 10: Push the code to the repository

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/p-4$ git push -u origin main
Username for 'https://github.com': A2003nu
Password for 'https://A2003nu@github.com':
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 8 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 724.33 KiB | 3.55 MiB/s, done.
Total 719 (delta 141), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (141/141), done.
To https://github.com/A2003nu/prog-4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Output:

The screenshot shows the GitHub interface for a repository named 'prog-4' (Public). The repository has 1 branch (main) and 0 tags. The commit history shows an initial commit by 'A200nu' 25 minutes ago, with a commit hash of 'af3b8e5'. The commit includes files: 'node_modules', 'Dockerfile', 'index.js', 'package-lock.json', and 'package.json'. Below the commit list, there is a section for the README, which currently says 'Add a README' and 'Help people interested in this repository understand your project by adding a README.' with a green 'Add a README' button.

prog-4 (Public)

main 1 Branch 0 Tags

Go to file Add file Code

A200nu Initial commit af3b8e5 · 25 minutes ago 1 Commit

node_modules	Initial commit	25 minutes ago
Dockerfile	Initial commit	25 minutes ago
index.js	Initial commit	25 minutes ago
package-lock.json	Initial commit	25 minutes ago
package.json	Initial commit	25 minutes ago

README

Add a README

Help people interested in this repository understand your project by adding a README.

Add a README